

Women Safety Alert System Code:

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
#include <TinyGPS++.h>
```

```
#include <HardwareSerial.h>
```

```
#define BUZZER 26
```

```
#define GSM_RX 16
```

```
#define GSM_TX 17
```

```
#define GPS_RX 4
```

```
#define GPS_TX 2
```

```
#define BUZZER_FREQUENCY 2000
```

```
#define BUZZER_RESOLUTION 8
```

```
#define BUZZER_DUTY 128
```

```
String phoneNumber = "+91XXXXXXXXXX";
```

```
#define LCD_ADDRESS 0x27
```

```
LiquidCrystal_I2C lcd(LCD_ADDRESS, 16, 2);
```

```
TinyGPSPlus gps;
HardwareSerial gsmModule(2);
HardwareSerial gpsModule(1);

String typedText = "";
bool alertDone = false;

void buzzerON() {
    ledcWrite(BUZZER, BUZZER_DUTY);
}

void buzzerOFF() {
    ledcWrite(BUZZER, 0);
}

void buzzerBeep(int times) {
    for (int i = 0; i < times; i++) {
        buzzerON();
        delay(300);
        buzzerOFF();
        delay(200);
    }
}
```

```
void setup() {  
  
    Serial.begin(115200);  
    delay(1000);  
    Serial.println("=====");  
    Serial.println("  WOMEN ALERT SYSTEM STARTING");  
    Serial.println("=====");  
  
    ledcAttach(BUZZER, BUZZER_FREQUENCY,  
BUZZER_RESOLUTION);  
    buzzerOFF();  
  
    Serial.println("[BUZZER] Testing... (you should hear 2 beeps)");  
    buzzerBeep(2);  
    Serial.println("[BUZZER] Test done.");  
  
    Serial.println("[LCD] Starting...");  
    Wire.begin(21, 22);  
    lcd.init();  
    lcd.backlight();  
    delay(500);  
  
    lcd.clear();  
    lcd.setCursor(0, 0);
```

```
lcd.print("Women Alert Sys");  
lcd.setCursor(0, 1);  
lcd.print("Starting...");  
Serial.println("[LCD] Started. Can you see text on LCD?");  
  
delay(2000);  
  
Serial.println("[GSM] Starting...");  
gsmModule.begin(9600, SERIAL_8N1, GSM_RX, GSM_TX);  
delay(3000);  
  
Serial.println("[GSM] Sending AT command...");  
gsmModule.println("AT");  
delay(1000);  
String gsmResponse = readGSM(2000);  
Serial.println("[GSM] Response: " + gsmResponse);  
  
if (gsmResponse.indexOf("OK") != -1) {  
    Serial.println("[GSM] GSM module is working!");  
} else {  
    Serial.println("[GSM] WARNING: GSM not responding. Check  
wiring and power.");  
}
```

```
gsmModule.println("ATE0");
```

```
delay(500);
```

```
readGSM(500);
```

```
Serial.println("[GSM] Checking SIM card...");
```

```
gsmModule.println("AT+CPIN?");
```

```
delay(1000);
```

```
String simStatus = readGSM(2000);
```

```
Serial.println("[GSM] SIM Status: " + simStatus);
```

```
if (simStatus.indexOf("READY") != -1) {
```

```
    Serial.println("[GSM] SIM card is OK!");
```

```
} else {
```

```
    Serial.println("[GSM] WARNING: SIM card problem. Check if  
SIM is inserted.");
```

```
}
```

```
Serial.println("[GSM] Checking network...");
```

```
gsmModule.println("AT+CREG?");
```

```
delay(1000);
```

```
String netStatus = readGSM(2000);
```

```
Serial.println("[GSM] Network: " + netStatus);
```

```
gsmModule.println("AT+CMGF=1");
```

```
delay(500);  
readGSM(500);  
Serial.println("[GSM] Setup complete.");
```

```
Serial.println("[GPS] Starting...");  
gpsModule.begin(9600, SERIAL_8N1, GPS_RX, GPS_TX);  
Serial.println("[GPS] Started. GPS needs open sky to get signal.");
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Type 'start' in");  
lcd.setCursor(0, 1);  
lcd.print("Serial Monitor");
```

```
Serial.println("");  
Serial.println("=====");  
Serial.println("    SYSTEM IS READY    ");  
Serial.println(" Type start + press Enter ");  
Serial.println("=====");  
}
```

```
void loop() {
```

```
    while (gpsModule.available()) {
```

```
gps.encode(gpsModule.read());
}

while (Serial.available()) {
    char letter = (char)Serial.read();

    if (letter == '\n' || letter == '\r') {

        typedText.trim();
        typedText.toLowerCase();

        if (typedText == "start" && alertDone == false) {
            alertDone = true;
            runAlertSystem();
        } else if (alertDone == false) {
            Serial.println("Type start and press Enter to trigger alert.");
        }

        typedText = "";

    } else {
        typedText += letter;
    }
}
```

```
}
```

```
void runAlertSystem() {
```

```
    Serial.println("");
```

```
    Serial.println("==== ALERT TRIGGERED! ====");
```

```
    buzzerON();
```

```
    Serial.println("[BUZZER] Buzzer is ON and buzzing loudly!");
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("System Initiated");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Please wait...");
```

```
    Serial.println("[LCD] System Initiated");
```

```
    delay(1500);
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("System is Ready!");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Getting GPS...");
```

```
    Serial.println("[LCD] System is Ready!");
```



```
Serial.println("[GPS] Getting live location...");
```

```
double myLatitude = 0.0;
```

```
double myLongitude = 0.0;
```

```
bool gotLocation = getGPSCoordinates(myLatitude, myLongitude);
```

```
if (gotLocation == true) {
```

```
    String mapLink = "https:
```

```
        + String(myLatitude, 6)
```

```
        + ","
```

```
        + String(myLongitude, 6);
```

```
    Serial.println("[GPS] Location found: " + mapLink);
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("Sending SMS...");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Please wait...");
```

```
    String smsText = "EMERGENCY ALERT!\n\n Need Immediate  
Assistance.\n\n live location:\n"
```

```
        + mapLink
```

```
        + "\n\n Please reach immediately.";
    sendSMS(smsText);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Alert SMS Sent!");
    lcd.setCursor(0, 1);
    lcd.print("Stay Safe!");
    Serial.println("[SMS] Alert sent successfully!");

} else {

    Serial.println("[GPS] Failed. Sending SMS without location.");

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("No GPS Signal");
    lcd.setCursor(0, 1);
    lcd.print("Sending SMS...");

    sendSMS("ALERT! She needs HELP! GPS location not available.
Please call her immediately!");

    lcd.clear();
```

```
    lcd.setCursor(0, 0);  
    lcd.print("Alert Sent!");  
    lcd.setCursor(0, 1);  
    lcd.print("(No GPS)");  
}
```

```
delay(2000);  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("ALERT ACTIVE");  
lcd.setCursor(0, 1);  
lcd.print("Buzzer is ON!");
```

```
Serial.println("");  
Serial.println("[BUZZER] Buzzer stays ON continuously.");  
Serial.println("[INFO] Press RESET button on ESP32 to restart  
system.");  
}
```

```
bool getGPSCoordinates(double &lat, double &lon) {
```

```
    Serial.println("[GPS] Waiting for GPS fix... (max 60 seconds)");  
    Serial.println("[GPS] NOTE: GPS needs open sky. Move outside if  
indoors.");
```

```
unsigned long startTime = millis();
unsigned long waitTime = 60000;

while (millis() - startTime < waitTime) {

    while (gpsModule.available()) {
        gpsModule.encode(gpsModule.read());
    }

    if (gps.location.isValid()) {
        lat = gps.location.lat();
        lon = gps.location.lng();
        Serial.println("[GPS] Fix found!");
        Serial.println("[GPS] Latitude: " + String(lat, 6));
        Serial.println("[GPS] Longitude: " + String(lon, 6));
        return true;
    }

    int secondsPassed = (millis() - startTime) / 1000;

    if (secondsPassed % 5 == 0) {
        Serial.println("[GPS] Still searching... " + String(secondsPassed)
            + " seconds passed.");
    }
}
```

```
}
```

```
lcd.clear();
```

```
lcd.setCursor(0, 0);
```

```
lcd.print("Getting GPS...");
```

```
lcd.setCursor(0, 1);
```

```
lcd.print("Wait: " + String(secondsPassed) + "s");
```

```
delay(1000);
```

```
}
```

```
Serial.println("[GPS] Timed out. No GPS fix after 60 seconds.");
```

```
return false;
```

```
}
```

```
void sendSMS(String message) {
```

```
Serial.println("[SMS] Sending to: " + phoneNumber);
```

```
Serial.println("[SMS] Message: " + message);
```

```
gsmModule.println("AT+CMGF=1");
```

```
delay(1000);
```

```
readGSM(1000);
```

```
gsmModule.print("AT+CMGS=\"");  
gsmModule.print(phoneNumber);  
gsmModule.println("\"");  
delay(2000);
```

```
String prompt = readGSM(2000);  
Serial.println("[SMS] GSM prompt: " + prompt);
```

```
if (prompt.indexOf(">") != -1) {
```

```
    Serial.println("[SMS] Typing message...");  
    gsmModule.print(message);  
    delay(500);
```

```
    gsmModule.write(26);  
    delay(5000);
```

```
    String result = readGSM(5000);  
    Serial.println("[SMS] Result: " + result);
```

```
    if (result.indexOf("+CMGS") != -1) {  
        Serial.println("[SMS] SMS SENT SUCCESSFULLY!");  
    } else {
```

```
    Serial.println("[SMS] SMS may have failed. Check SIM balance  
and signal.");
```

```
    }
```

```
  } else {
```

```
    Serial.println("[SMS] ERROR: GSM did not respond with >  
prompt.");
```

```
    Serial.println("[SMS] Check: 1) GSM power 2) SIM card 3) GSM  
wiring");
```

```
  }
```

```
}
```

```
String readGSM(unsigned long waitMs) {
```

```
  String response = "";
```

```
  unsigned long start = millis();
```

```
  while (millis() - start < waitMs) {
```

```
    while (gsmModule.available()) {
```

```
      response += (char)gsmModule.read();
```

```
    }
```

```
  }
```

```
  response.trim();
```

```
  return response;
```

```
}
```